Mnjoli Water-Stress Research Record

Data, methods, results, deployment, and research limitations

STUDY AREA Mnjoli W60E catchment, Eswatini	**RECORD VERSION** Version 1.0 | 7 November 2026
RESEARCH SCOPE Hydroclimatic index, model comparisons, drought-onset analysis, input-product sensitivity	**DEPLOYED PROTOTYPE** Satellite surface-area and weather-ensemble indicators; no storage forecast

Important interpretation

The live system does not predict reservoir storage, water supply, or drought impacts. The reported historical model scores target a standardized hydroclimatic index, not dam operations. Operational storage observations sufficient for calibration and validation have not been obtained.

Prepared as a research record for manuscript preparation. This document is not peer reviewed.

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Scope and interpretation. *This document records several related but distinct research experiments and the current remote-data prototype. It is not a claim that the system forecasts reservoir storage, water supply, or drought impacts. The live dashboard currently reports a satellite-derived surface-area proxy and a weather-ensemble hydroclimate context as separate indicators. Operational storage observations sufficient for calibration and validation have not been obtained.*

1. Executive summary

The project began as a machine-learning study of monthly hydroclimatic water stress in Eswatini. It used remotely sensed precipitation and reanalysis-derived meteorological and land-surface variables to predict a standardized water balance one month ahead. The research was subsequently broadened to examine drought-onset warnings and the reliability of historical rainfall inputs. Because there is no verified, timestamped Mnjoli storage series, inflow or release history, or stage-area-volume curve in the project data, the deployed Mnjoli Water-Risk Watch does not estimate reservoir storage. It presents:

1. Sentinel-2 visible reservoir surface area inside an informal mapped water

polygon; and

2. a 30-day Open-Meteo seasonal ensemble summary for precipitation, reference

evapotranspiration (ET0), and precipitation minus ET0.

Those indicators are explicitly not interpreted as reservoir volume, inflow, or service-risk probabilities.

The expanded historical water-balance index is reproducible over 1981-2025. However, drought-onset events are rare (30 onsets in 540 months), and retrospective warning experiments show a sensitivity-versus-warning-burden trade-off. The most recent one-month warning-budget screen detected 3 of 13 onsets in pooled, previously examined development folds. This is not sufficient evidence for an operational alert. A separate 2020-2025 continuous-anomaly test showed better average error for a regularized model than persistence, but contained only one month below the drought threshold; it cannot establish drought-detection skill.

A CHIRPS v2-to-v3 overlap study found substantial historical product differences. On the shared 1981-2025 period, v3 precipitation total was 19.23% higher than the existing v2 catchment series. A full-period matched-reducer audit found the v2 spatial-reducer difference was small compared with the v2/v3 difference. Nevertheless, v3 processing, daily disaggregation, and mirror rounding remain possible contributors, so this is not a pure causal attribution to product version alone.

The water-risk API and Firebase-hosted dashboard are deployed. Live endpoint checks returned weather ensemble data and a Sentinel-2 surface-area estimate. These are operational checks of data retrieval and display, not hydrological validation.

2. Research questions and project evolution

The original research question was whether environmental conditions observed at month *t* could predict a standardized hydroclimatic water-stress index at month *t+1*. The work emphasized chronological validation, leakage prevention, regularization, and comparison against simple baselines.

The project then explored a second question: can a model issue useful warnings before a month crosses a drought threshold? Continuous index regression and binary onset classification were treated as different targets, with different metrics and limitations.

The intended application later shifted toward water availability at Mnjoli and a useful outlook up to 30 days ahead. Source review did not identify reliable, dated reservoir-operation records. The product was therefore deliberately reframed as a **relative satellite-and-weather proxy**, rather than a synthetic storage forecast. The user-facing prototype keeps the satellite observation and seasonal weather context separate.

These phases are not one uninterrupted model experiment. In particular:

- The original short-record Ridge test, expanded historical continuous-index

work, onset-warning backtests, CHIRPS transition study, and live proxy use different targets, inputs, periods, and evaluation designs.

- Results from one phase should not be combined into a single headline

performance estimate.

- Deployment of an API or dashboard demonstrates software integration, not

predictive validity.

3. Study area and data

The focal geography is the Mnjoli W60E catchment in Eswatini. Historical catchment products use a HydroSHEDS basin polygon. The satellite prototype uses a separate OpenStreetMap water polygon for Mnjoli Dam. The reservoir water polygon is an informal mapped geometry, not a surveyed maximum-pool boundary, and it must not be used to infer storage volume without additional hydraulic information.

Data or product	Role	Coverage / resolution	Important qualification
CHIRPS v2 daily (`UCSB-CHG/CHIRPS/DAILY`)	Historical precipitation for the existing expanded index and live ingestion workflow	0.05-degree grid; historical archive 1981-2025, refreshed through 2026-09-30 in the audited snapshot	Delayed and revisable; retrieved series does not by itself establish what was available at forecast issue time
ERA5-Land daily aggregate	PET and meteorological / land-surface predictors	Historical archive through 2025; refreshed through 2026-10-26 in the audited snapshot	Common complete cutoff with CHIRPS was 2026-09-30 at the November 2026 refresh
CHIRPS v3 final/rnl mirror	Version-transition sensitivity analysis	0.05-degree grid; 1981-01 through 2025-12	Public dynamical.org processed mirror; daily values are disaggregated and rounded; final release is typically about 2-3 weeks after month end
Sentinel-2 Surface Reflectance Harmonized	Recent satellite surface-water extent proxy	Scene-based imagery; 20 m processing scale for the prototype's area reductions	Cloud/shadow masking, pixel classification, shoreline effects, and mapped-boundary uncertainty affect estimates
Open-Meteo seasonal forecast (ECMWF EC46 context)	30-day weather ensemble	About 36 km model-grid scale; 50 paired precipitation and ET0 members in the observed live response	Coarse spatial representation and extended-range uncertainty; not a catchment runoff or reservoir-inflow forecast
OpenStreetMap relation 1792362	Mnjoli Dam water polygon for satellite processing	Polygon geometry; 908 vertices in the retrieved relation	Informal mapped water boundary; attribution to OpenStreetMap contributors required
EWSC public dam-level display	Investigated as a potential storage observation	Public homepage display sampled during the research	“102%” had no observation timestamp, percentage definition, or usable history; repeated archived display is not validation ground truth

3.1 Historical refresh and issue-time freshness

The audited shadow-data refresh retrieved CHIRPS through 30 September 2026 and ERA5-Land through 26 October 2026. Their common complete daily cutoff was 30 September 2026, 38 days before the 7 November refresh. The refreshed monthly series covered 1981-01 through 2026-09 (549 months), preserved the existing 540 historical months through 2025 within tolerance, and passed its data audit.

Two retrieval snapshots on 6 and 7 November had the same common cutoff and identical overlapping source values. This only describes those sampled retrievals. It does not establish a general first-publication delay or a revision rate. A separate October-target shadow forecast was created after October had started and is documented as a pipeline exercise, not timely forecast evidence.

4. Index construction and label definitions

For the expanded historical analysis, monthly climatic water balance is:

```
WB_t = P_t - PET_t
```

where `P_t` is catchment precipitation and `PET_t` is potential evapotranspiration. The three-month balance is the rolling sum of monthly water balance:

```
WB3_t = WB_t + WB_(t-1) + WB_(t-2)
```

The standardized water-balance anomaly is calculated separately for each calendar month using the 1981-2010 reference period:

```
SWBA_t = (WB3_t - mean_calendar_month,1981-2010) / sample_SD_calendar_month,1981-2010
```

The reference standard deviation uses sample standard deviation (`ddof=1`). Negative values denote drier-than-reference hydroclimatic conditions; positive values denote wetter-than-reference conditions. SWBA is a standardized climatic anomaly, not a probability, reservoir fullness percentage, or direct water-supply measure.

For onset classification:

- A drought month is `SWBA <= -1.0`.
- An onset is the first drought month following an immediately preceding valid

month with `SWBA > -1.0`.

- A forecast origin is eligible only when its origin-month SWBA is greater

than -1.0, so the task is advance warning rather than detecting an already active drought.

- The primary warning horizon is one month. Two- and three-month analyses are

exploratory.

The original short-record Ridge study used a distinct, training-period climatology and a 2024-2025 independent test. Those values must not be merged with expanded SWBA results using the fixed 1981-2010 reference.

5. Historical data and label audit

The expanded monthly dataset contains 540 consecutive months from January 1981 through December 2025. It has two missing SWBA / three-month-balance rows at the beginning of the rolling window and no missing precipitation, PET, or first-layer soil-moisture predictor rows. The audit reconstructed monthly balance, rolling three-month sums, reference climatology, and SWBA from source fields.

Audit check	Maximum absolute difference	Outcome
Monthly balance vs precipitation minus PET	8.97e-14	Passed
Three-month balance vs rolling sum	2.27e-13	Passed
Stored calendar-month reference mean	1.14e-13	Passed
Stored calendar-month sample standard deviation	4.26e-14	Passed
Stored vs reconstructed SWBA	5.44e-15	Passed

The audit found 77 drought months and 30 onset events. Eligible samples with complete predictors numbered 460, 459, and 458 for one-, two-, and three-month leads respectively. Onset counts are small and uneven across chronological blocks; the one-month test blocks contain 5, 5, 2, and 1 events for 2006-2010, 2011-2015, 2016-2020, and 2021-2025.

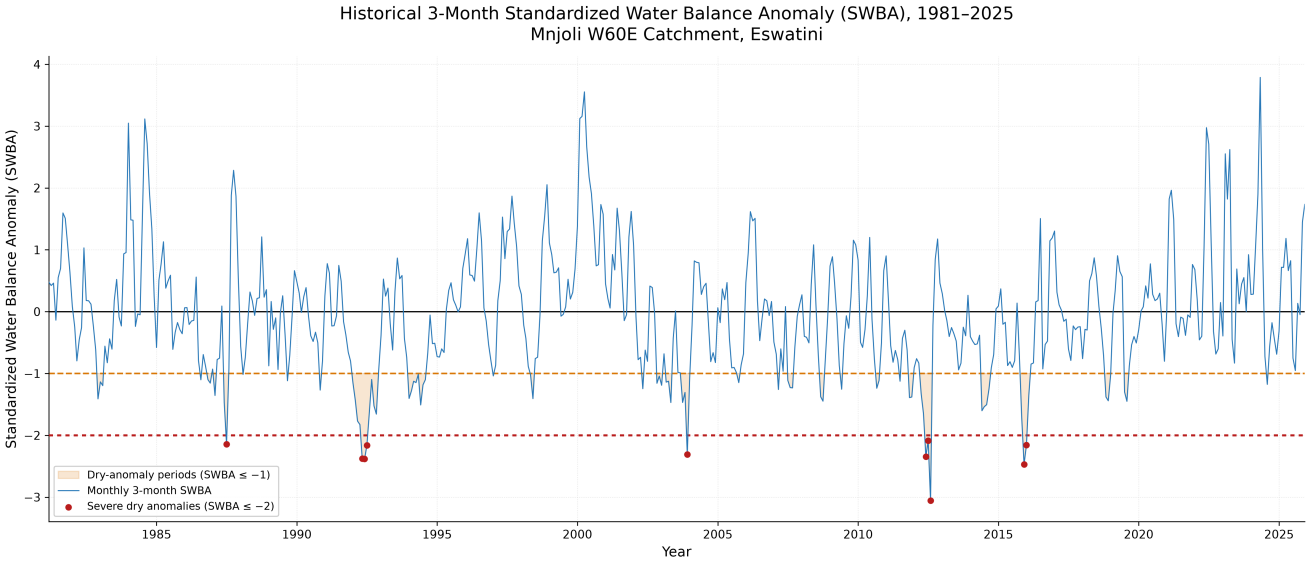


Figure 1. Historical three-month standardized water-balance anomaly (SWBA), 1981?2025. The dashed line at ?1 is the study drought-month threshold; the plot is an index history, not a reservoir record.

6. Modeling experiments and results

6.1 Original short-record continuous forecast

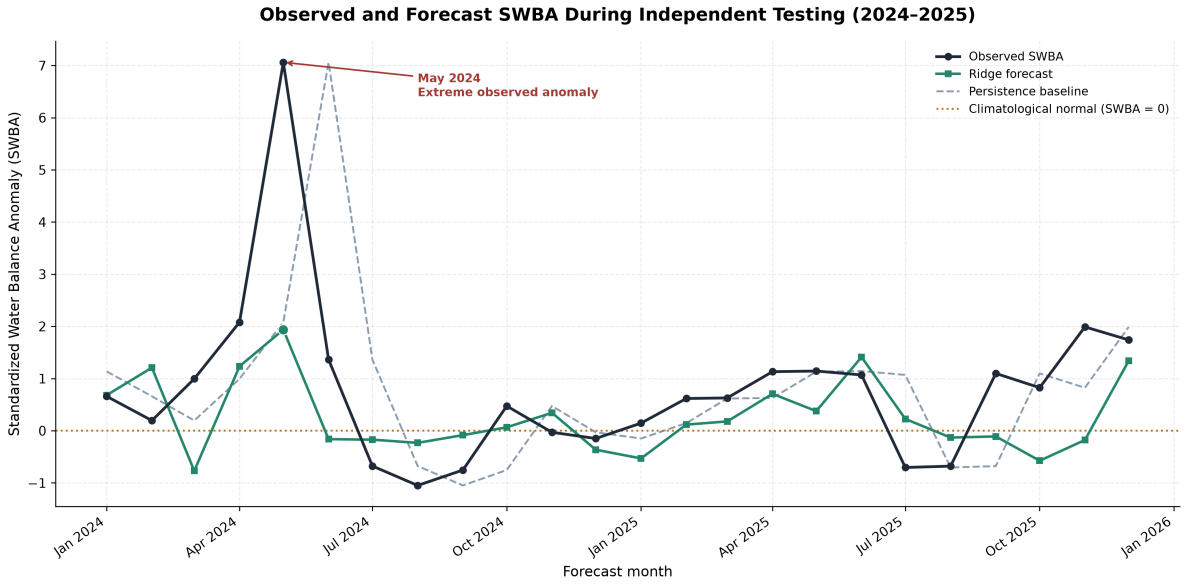
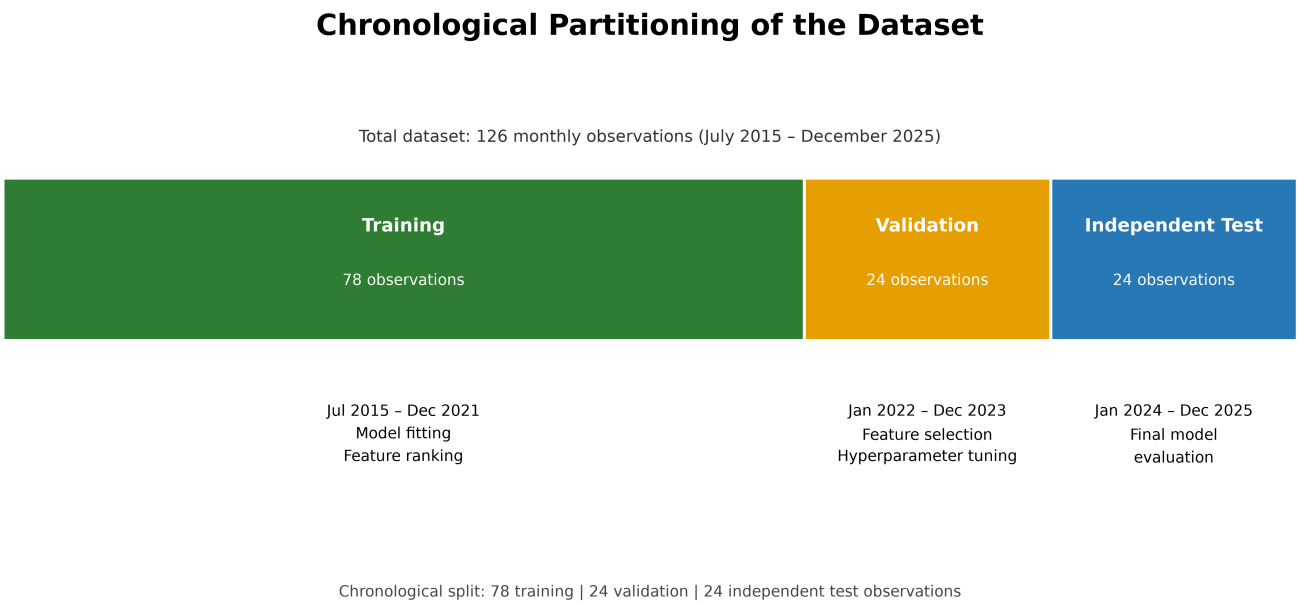


Figure 4. Observed and one-month-ahead Ridge SWBA estimates over the 2024?2025 independent test. The figure belongs to the original short-record experiment and should be interpreted with its small sample and extreme-anomaly limitations.

The repository documents a leakage-aware Ridge Regression pipeline with median imputation, standardized predictors, and Ridge alpha 0.05. Its target is a one-month-ahead continuous stress index. Feature ranking and model configuration were selected without random temporal shuffling; the 24-observation independent test covers January 2024 through December 2025.

Test result	Ridge
MAE	0.9625
RMSE	1.3906
R-squared	0.2124
Test RMSE: persistence	1.7568
Test RMSE: climatology	1.7589

Figure 2. Chronological partition of the original Ridge study



The original short-record experiment used 78 monthly observations for training (July 2015?December 2021), 24 for validation (January 2022?December 2023), and 24 in the independent test (January 2024?December 2025). The figure describes that experiment only; it is not the expanded SWBA study split.

The README reports an approximately 20.84% RMSE reduction versus persistence and 20.93% versus climatology on that specific test. Only one test observation was at or below SWBA -1.0, and none was at or below -1.5 or -2.0. Two extreme absolute anomalies in this test were both positive, not drought events. Therefore this test is not meaningful evidence of dry-event or drought-warning skill.

6.2 Expanded continuous-anomaly experiment

A later expanded analysis evaluated January 2020 through December 2025 using a longer historical series and a different fixed-reference SWBA definition. Reported final-test metrics were:

Model / benchmark	MAE	RMSE	R-squared
Lasso	0.5089	0.7001	0.5155
Ridge	0.5077	0.7123	0.4985
Tuned XGBoost	0.5420	0.7465	0.4491
Persistence	0.6952	0.9833	0.0443
Seasonal climatology	0.7970	1.1294	-0.2608
Training-mean baseline	0.8040	1.1435	-0.2924

Rolling-origin validation over the reported folds favored an ElasticNet/Lasso configuration on average (mean RMSE 0.5253, mean R-squared 0.5804), with Ridge close behind (mean RMSE 0.5304, mean R-squared 0.5710). These are continuous-target results on the expanded SWBA and do not demonstrate forecasting of reservoir level, storage, inflow, or service impacts. The 2020-2025 holdout contained only one SWBA drought month at or below -1.0, so continuous average scores must not be recast as drought detection performance.

6.3 Drought-onset warning experiments

The onset task was audited and evaluated separately. Prior exploratory model searches were replaced by chronology-aware screens and a warning-budget feasibility analysis. Previously examined 2006-2025 folds are development backtests, not an untouched final test. Thresholds were selected on a prior five-year validation block for each outer test block, under user-selected limits of no more than 0.5 false-warning episodes per year and 3 false-warning months per year.

In the November 2026 pooled one-month warning-budget screen, the fixed four-feature hydroclimate logistic model:

- detected 3 of 13 unique onset events (event recall 0.231);
- produced 7 false-warning months and 5 false-warning episodes over four

five-year test blocks; and

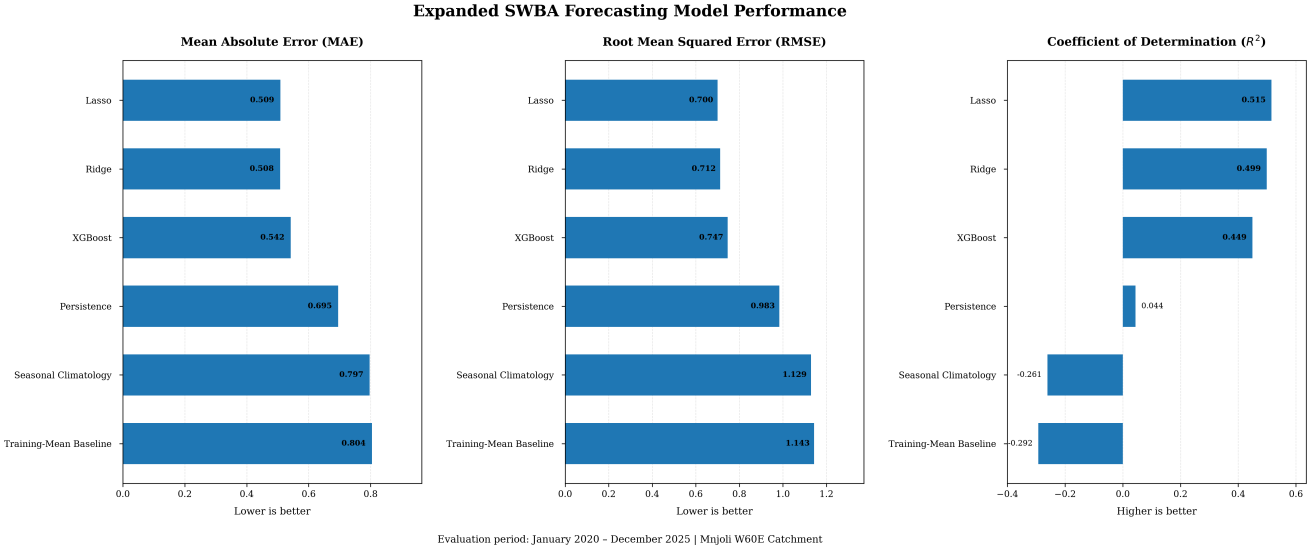
- met both prespecified warning limits in all four historical folds.

The seasonal and training-prevalence baselines produced no warnings and detected no onset in the one-month pooled test. Meeting the false-warning limits by issuing few warnings is not sufficient if event sensitivity is low. At two months, the hydroclimate candidate detected 8 of 13 events but met both limits in only 2 of 4 folds (11 false-warning months and 10 episodes). At three months, it detected 11 of 13 but met both limits in 2 of 4 folds (16 false-warning months and 12 episodes). Overlapping forecast windows make origin-level outcomes dependent; an onset is counted once for event recall even if several warnings cover it.

A separate nested one-month backtest artifact reports a selected-model summary of 3/13 onset hits, 12 false-warning months, and 5 false-warning episodes across 209 test samples; its Brier score was 0.0553 versus 0.0589 for its training-prevalence baseline. This is a different experiment and threshold-selection artifact than the fixed-method warning-budget screen. Both are retrospective, previously examined development results and should be reported as exploratory, not as independent confirmation.

The partial 2026 holdout covered January through September and contained zero observed onsets. Its Brier score was 0.00054 against 0.00523 for the prevalence baseline, but with zero events it cannot estimate onset recall or detection skill. Zero warnings in that partial period likewise cannot establish a long-run warning burden.

Figure 3. Expanded continuous-SWBA model comparison



Final-test comparison for January 2020?December 2025. The evaluated target is the standardized water-balance anomaly (SWBA), not reservoir level, storage, inflow, or water supply. The accompanying research record provides values, baselines, sample limitations, and validation context.

6.4 CHIRPS v2-to-v3 comparison

The comparison used complete catchment-month coverage for January 1981 through December 2025 (540 months). CHIRPS v3 final/rnl daily fields were obtained from the dynamical.org Icechunk/Zarr mirror of the CHC product; existing ERA5-Land PET was held fixed. V3 was evaluated with the existing v2 1981-2010 reference statistics and again after recalculating the reference statistics from v3. This separates reference-period effects from the precipitation series change.

Comparison	Result
Existing v2 total precipitation	31,836.12 mm
V3 total precipitation	37,958.46 mm
V3 minus v2 total	+6,122.34 mm (+19.23%)
Monthly total correlation	0.94694
Monthly mean bias (v3 minus v2)	+11.34 mm
Monthly MAE / RMSE	15.16 / 24.90 mm
Drought months, v2	77
Drought months, v3 using v2 reference	54
Drought months, v3 with its own 1981-2010 reference	75
Onsets, v2 / v3 with v2 reference / v3 own reference	30 / 25 / 32

With the v2 reference retained, 27 drought-month labels changed; with v3's own reference, 24 labels changed. Those are changes to historical index labels, not changes in model skill. The data came from a processed mirror that rounds stored values and documents ERA5-based daily disaggregation of pentadal CHIRPS totals. Thus the differences cannot be attributed solely to the version-number change.

A full-period reducer-equivalence audit sampled all 11 native v2 cells intersecting the catchment for 16,436 days (180,796 pixel-day samples), then applied the same fractional-area weighting method used for v3. The weighted v2 total was 31,802.34 mm versus 31,836.12 mm for the archived Earth Engine mean reducer: -33.77 mm (-0.106%) over 45 years. Daily correlation was 0.999989, daily MAE 0.00860 mm, and maximum annual-total difference 0.218%. Under this matched spatial reducer, v3 remained 19.36% above v2. This indicates that the reducer difference is small relative to the observed precipitation-product difference, while not resolving mirror processing, daily disaggregation, or rounding as explanations.

V3 final data are not used in the live model. Its reported typical final release delay of two to three weeks after month-end also does not by itself meet an early-warning freshness requirement.

7. Deployed relative water-availability prototype

7.1 Satellite surface-area method

The backend queries the Google Earth Engine collection `COPERNICUS/S2_SR_HARMONIZED` over the Mnjoli OSM water polygon and a 90-day lookback, limited to 100 images. It computes the modified normalized difference water index using green (B3) and shortwave infrared (B11) reflectance:

```
MNDWI = (B3 - B11) / (B3 + B11)
```

Pixels with MNDWI greater than zero are classified as water. Sentinel scene-classification pixels 4, 5, and 6 are treated as valid; clouds, shadows, snow, and unclassified pixels are excluded. Water and valid areas are calculated using pixel area at a 20 m reduction scale. A scene is retained only when valid pixels cover at least 80% of the mapped polygon. Multiple images on the same acquisition date are deduplicated, keeping the observation with better valid coverage. The service returns the two most recent qualifying dates and their relative area change when available.

This is a spectral surface-extent proxy. It is not a water-level measurement or storage-volume conversion. It depends on the OSM polygon, valid-pixel coverage, classification threshold, atmospheric/surface conditions, and shoreline-pixel behavior. No local reference labels or surveyed stage-area curve were available to calibrate the threshold or validate area accuracy.

7.2 Weather-ensemble method

The service requests 30 daily precipitation and reference ET0 values from the Open-Meteo seasonal forecast API at the Mnjoli-area requested coordinates. Precipitation and ET0 are paired by ensemble-member identifier. For each complete member, daily values are summed over the 30-day period, and precipitation minus ET0 is calculated using the paired member totals. The API returns the P10, median, P90, and arithmetic ensemble mean. Fewer than 10 complete paired members or invalid 30-day series are rejected.

Gross rainfall volume is calculated as catchment precipitation depth multiplied by the 896.9 km2 catchment area. Since one millimetre over one square kilometre equals one megalitre, the output unit is ML. This is gross rainfall before interception, infiltration, runoff generation, routing, reservoir operations, or withdrawals. `P - ET0` is a hydroclimate indicator, not runoff, reservoir inflow, storage change, or water supply.

7.3 Live snapshot (retrieval: 7 November 2026)

The public API was directly checked after deployment. The following values describe that retrieval snapshot; they are not an archived hindcast or a verified observation series.

Live indicator	Snapshot result
Satellite acquisition date	2026-10-31
Estimated surface area	12.656 km2
Valid-pixel coverage	92.46%
Previous qualifying scene	2026-10-19
Previous area / coverage	13.643 km2 / 99.37%
Relative area change	-7.24%
Weather period	2026-11-07 to 2026-12-06
Weather ensemble members	50
Precipitation total, P10 / median / P90	54.76 / 98.65 / 155.68 mm
Reference ET0 total, median	141.06 mm
P - ET0, P10 / median / P90	-108.56 / -43.94 / +32.83 mm
Gross catchment rainfall, median	88,479 ML
Resolved model grid point	-25.9953, 31.8103; elevation 308 m

The relative surface-area change must not be phrased as a 7.24% storage decrease. The seasonal ensemble median climate balance being negative does not mean the reservoir will lose a corresponding volume; there is no calibrated catchment-to-reservoir response model.

7.4 API and user interface

The FastAPI application exposes:

- `GET /water-risk/climate-balance`
- `GET /water-risk/surface-area`

The frontend dashboard is hosted at <https://eswatiniwaterstress.web.app/water-risk>. On 7 November 2026, both endpoints returned HTTP 200 and the public dashboard displayed the dated satellite and forecast indicators. A deployment bug initially created an Earth Engine geometry before initializing the Earth Engine client. The initialization order was corrected and regression-tested; the endpoint subsequently returned a valid scene summary. This incident is an implementation-history note, not a data-quality result.

The dashboard links and response payloads describe source and retrieval time, but a persistent time-series archive of repeated satellite and forecast snapshots still needs to be established for retrospective verification.

8. Evidence-based interpretation

The evidence supports three limited conclusions:

1. A reproducible monthly hydroclimatic anomaly series can be constructed

from the archived precipitation and PET products, and a continuous one-month-ahead model can outperform simple persistence/climatology baselines on some historical test designs.

2. The drought-onset problem is event-sparse. Retrospective results show that

warning-budget compliance can coexist with low onset sensitivity, and apparent sensitivity rises at longer horizons with increased warning burden. The evidence does not justify operational drought alerts.

3. Remote satellite/weather services can populate a useful, clearly limited

monitoring interface. They currently provide independent contextual indicators, not a validated water-availability or storage forecast.

The evidence does **not** show that the dashboard predicts water shortage, reservoir storage, service interruption, or operator action. Nor does an API response establish the physical accuracy of MNDWI-derived area or seasonal forecast skill over the catchment.

9. Limitations and unresolved research requirements

Hydrological and operational data

- No verified dated Mnjoli storage or level observations, inflow, release,

downstream discharge, abstraction, demand, or service-impact time series were identified in the project data.

- The public 102% value lacks timestamp and definition, and sampled archives

repeat it. It is not a calibration target.

- There is no surveyed reservoir stage-area-volume relation for converting

image-derived area to level or storage.

- No local rain-gauge series or independent satellite-area reference labels

were used for validation.

Forecast and remote-sensing uncertainty

- The 30-day seasonal weather product is coarse-grid extended-range guidance,

not a locally calibrated catchment forecast.

- P - ET0 omits soil storage, runoff generation, routing, groundwater,

reservoir operation, and demand.

- Satellite results are sensitive to cloud masking, valid coverage, water

threshold, shoreline mix, scene availability, and polygon quality.

- The mapped OSM water polygon is not a surveyed maximum extent.
- Live responses observed on one retrieval date do not provide temporal

accuracy, forecast reliability, or trend significance.

Statistical evidence

- The drought-onset series has only 30 historical events, unevenly distributed

in time; outer test blocks contain very few events.

- Development folds have already influenced model experimentation and are not

independent validation.

- Overlapping lead windows produce dependent samples and warnings.
- The 2026 partial holdout has no onset, so detection skill cannot be estimated.

- Input retrieval freshness and source vintages do not yet establish exact

issue-time availability or general source-revision behavior.

- Results from the 2015-2021/2024-2025 original Ridge experiment and expanded

1981-2025-reference experiments use different targets/reference designs and must be reported separately.

10. Recommended next research steps

1. **Archive each feed.** At fixed retrieval intervals, persist raw API

payloads, retrieval timestamps, source version/model, issue and valid dates, resolved coordinates, and hashes. Keep each forecast vintage immutable.

2. **Obtain operator observations.** Request dated level/storage, inflow,

releases, operating rules, capacity/datum definitions, quality flags, service impacts, and permission from the responsible data steward.

3. **Validate satellite area.** Compare Sentinel-derived area with

operator-approved observations or independently interpreted imagery; examine threshold, coverage, season, and shoreline sensitivity.

4. **Build a physical response model only when supported.** Link rainfall,

catchment response, reservoir inflow, evaporation, releases, and demand with unit-checked data. Do not equate gross rainfall with inflow.

5. **Lock target and decision use.** Operators should define what outcome

matters, the acceptable lead time, false-warning burden, and missed-event cost before selecting alert thresholds.

6. **Prospective non-alerting evaluation.** Freeze forecasts and inputs before

outcome windows begin. Accumulate enough independent events and report calibration, discrimination, warning episodes, lead time, and uncertainty.

7. **Keep product transitions explicit.** Do not silently replace v2 with

v3 or retrain on relabeled history. First archive vintages, compare processing, and assess downstream model/label sensitivity.

8. **Separate claims.** Present continuous hydroclimatic forecasting,

onset-warning classification, satellite monitoring, and reservoir storage/supply forecasting as separate research claims with their own validation.

11. Reproducibility map

Repository sources of truth and evidence used to prepare this record:

- `README.md` — original monthly Ridge study, model pipeline, 24-month test,

and application description.

- `documentation/DROUGHT_ONSET_EVALUATION_PROTOCOL.md` — index/event

definitions, audit gate, warning-budget methodology, CHIRPS transition, and shadow-evaluation limitations.

- `documentation/MNJOLI_WATER_RISK_WATCH_PLAN.md` — source inventory,

storage-data gap, product framing, and operational recommendations.

- `scripts/create_expanded_swba.py` — expanded SWBA construction.
- `scripts/audit_drought_onset_data.py` — historical index and label

reconstruction checks.

- `scripts/evaluate_warning_budget_feasibility.py` and

`scripts/run_nested_drought_onset_backtest.py` — chronological onset evaluation workflows.

- `scripts/run_chirps_v3_transition_comparison.py` and

`scripts/audit_chirps_v2_spatial_reducer.py` — CHIRPS product/reducer sensitivity analyses.

- `backend/water_availability.py` — live weather and Sentinel-2 data

processing.

- `data/external/mnjoli_reservoir_boundary.geojson` — OSM dam-water geometry

with attribution metadata.

- `outputs/expanded/drought_onset_model_preparation/data_audit/` — audited

1981-2025 SWBA and onset records.

- `outputs/expanded/drought_onset_model_preparation/warning_budget_feasibility_20261107T011001Z/`

— pooled and fold-level warning-budget outputs.

- `outputs/expanded/shadow_evaluation_data/refresh_through_2026-09/` —

refreshed data, manifest, and partial-2026 holdout evaluation.

- `outputs/expanded/chirps_v3_transition/run_through_2025_20261106T181031Z/`

— v2/v3 comparison and full-period reducer audit.

- `outputs/expanded/final_evaluation/final_test_metrics.csv` — expanded

continuous-anomaly test metrics.

- `outputs/expanded/rolling_origin_validation/rolling_origin_validation_summary.csv`

— continuous-target rolling-origin results.

Generated CSVs and manifests should be preserved with their recorded input hashes. When reporting any result, state the target/index definition, data product and vintage, period, split design, sample/event counts, and whether the evaluation was used during model development.

12. Candidate source citations

These are data-product and service references, not a complete scholarly bibliography. Check current product documentation, access dates, licenses, versions, and citation requirements before manuscript submission.

1. Funk, C. et al. (2026). CHIRPS v3.0. *Scientific Data*, 13, 718.

<https://doi.org/10.15780/G2JQ0P>

2. UCSB Climate Hazards Center. CHIRPS data archive and documentation.

<https://data.chc.ucsb.edu/products/CHIRPS/>

3. ECMWF. ERA5-Land hourly data documentation.

<https://www.ecmwf.int/en/era5-land>

4. Google Earth Engine Data Catalog. COPERNICUS/S2_SR_HARMONIZED.

<https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED>

5. Open-Meteo. Seasonal forecast API documentation.

<https://open-meteo.com/en/docs/seasonal-forecast-api>

6. OpenStreetMap contributors. Mnjoli Dam relation 1792362.

<https://www.openstreetmap.org/relation/1792362>

7. dynamical.org. UCSB-CHC CHIRPS Analysis Final mirror, snapshot

T5RTYW439H0E8VF37CQG; processed dataset license stated as CC BY 4.0. <https://dynamical.org/catalog/ucsb-chc-chirps-analysis-final/>

Suggested manuscript framing: “A data-constrained hydroclimatic monitoring and forecasting study for Mnjoli, Eswatini: index evaluation, input-product sensitivity, and a satellite/weather proxy prototype.” Avoid “validated reservoir storage forecast” or “operational drought-warning model” unless and until the corresponding observations and prospective evaluation exist.